

## TYPES OF CHEMICAL REACTIONS MCQ

1. Chemical reaction involves

- a) physical change only
- b) rearrangement of atoms forming new substances
- c) no bond breaking
- d) conservation of mass violation

Correct answer: b

2. During chemical reaction bonds are

- a) only formed
- b) only broken
- c) broken and new formed
- d) neither broken nor formed

Correct answer: c

3. Bond breaking process is

- a) exothermic
- b) endothermic
- c) neutral
- d) catalytic

Correct answer: b

4. Evolution of gas characteristic observed in

- a) precipitation only
- b) combination reactions
- c) many decomposition reactions
- d) neutralization only

Correct answer: c

5. Precipitate formation typical of

- a) combination
- b) decomposition
- c) double displacement
- d) combustion

Correct answer: c

6. Chemical equation reactants written on

- a) right side
- b) left side
- c) both sides
- d) arrow symbol

Correct answer: b

7. Balanced equation maintains conservation of

- a) volume
- b) atoms
- c) molecules
- d) energy only

Correct answer: b

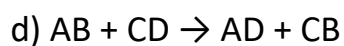
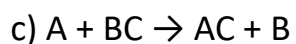
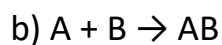
8.  $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$  balanced because atoms equal on

- a) left only
- b) right only
- c) both sides
- d) neither side

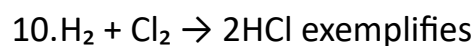
Correct answer: c

9. Combination reaction general form

- a)  $\text{AB} \rightarrow \text{A} + \text{B}$



Correct answer: b



a) decomposition

b) combination

c) displacement

d) double displacement

Correct answer: b



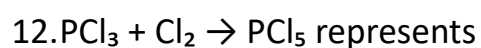
a) two compounds

b) metal non-metal

c) two non-metals

d) compound element

Correct answer: b



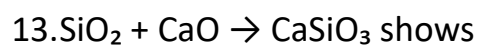
a) element element

b) compound element

c) compound compound

d) acid base

Correct answer: b



a) two elements

b) two compounds

c) element compound

d) three compounds

Correct answer: b

14. Most combination reactions nature

a) endothermic

b) exothermic

c) neutral

d) reversible

Correct answer: b

15. Decomposition requires energy as

a) heat light

b) electricity only

c) catalyst

d) pressure

Correct answer: a

16.  $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$  thermal decomposition by

a) electricity

b) light

c) heat

d) catalyst

Correct answer: c

17.  $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$  electrolytic decomposition using

a) sunlight

b) heat

c) electric current

d) chemical catalyst

Correct answer: c

18.  $2\text{AgCl} \rightarrow 2\text{Ag} + \text{Cl}_2$  photo decomposition by

- a) heat
- b) electricity
- c) light
- d) pressure

Correct answer: c

19. Combination synthesis opposite

- a) displacement
- b) decomposition
- c) double displacement
- d) combustion

Correct answer: b

20. Single displacement more active element displaces

- a) more active
- b) less active
- c) equal active
- d) inactive

Correct answer: b

21.  $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$  iron displaces copper because

- a) less reactive
- b) more reactive
- c) same reactivity
- d) catalyst present

Correct answer: b

22. Activity series determines displacement possibility based on

- a) atomic mass

- b) reactivity
- c) density
- d) melting point

Correct answer: b

23. Double displacement involves

- a) ion exchange
- b) atom rearrangement
- c) bond breaking only
- d) gas evolution

Correct answer: a

24.  $\text{Na}_2\text{SO}_4 + \text{BaCl}_2 \rightarrow \text{BaSO}_4 + 2\text{NaCl}$  produces

- a) gas
- b) precipitate
- c) heat
- d) light

Correct answer: b

25. Precipitation reaction product

- a) soluble gas
- b) insoluble solid
- c) aqueous solution
- d) volatile liquid

Correct answer: b

26. Neutralization acid base produces

- a) salt water
- b) precipitate
- c) gas

d) new element

Correct answer: a

27.  $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$  pH of products

a) acidic

b) basic

c) neutral

d) depends on amounts

Correct answer: c

28. Combustion requires

a) catalyst

b) oxygen oxidiser

c) electricity

d) light

Correct answer: b

29.  $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$  combustion of

a) carbon

b) hydrogen

c) methane

d) all above

Correct answer: d

30. Reversible reaction represented by

a) single arrow

b) double arrow

c) circle

d) equals sign

Correct answer: b

31. Haber process  $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$  example of

- a) irreversible
- b) reversible
- c) decomposition
- d) combustion

Correct answer: b

32. Coal burning irreversible because

- a) reversible conditions exist
- b) products cannot reform reactants
- c) catalyst present
- d) equilibrium reached

Correct answer: b

33. Reaction rate measures change in concentration per

- a) volume
- b) mass
- c) time
- d) temperature

Correct answer: c

34. Wood burning faster than iron rusting means

- a) higher activation energy
- b) higher reaction rate
- c) lower temperature
- d) more catalyst

Correct answer: b

35. Rate =  $-\text{d}[\text{A}]/\text{dt}$  negative sign indicates

- a) reactant concentration decreases

- b) product increases
- c) catalyst used
- d) equilibrium reached

Correct answer: a

36. Reaction rate increases with

- a) reactant concentration decrease
- b) temperature decrease
- c) catalyst absence
- d) large particle size

Correct answer: b

37. Collision theory explains rate depends on

- a) successful collisions
- b) total collisions
- c) temperature only
- d) pressure only

Correct answer: a

38. Activation energy required for

- a) all reactions
- b) exothermic only
- c) endothermic only
- d) equilibrium

Correct answer: a

39. Catalyst increases rate by lowering

- a) temperature
- b) concentration
- c) activation energy

d) pressure

Correct answer: c

40. Smaller particle size increases rate due to

a) less surface area

b) more surface area

c) higher density

d) lower temperature

Correct answer: b

41. Chemical equilibrium forward backward rates

a) different

b) equal

c) zero

d) increasing

Correct answer: b

42. Dynamic equilibrium concentrations

a) changing constantly

b) constant over time

c) increasing

d) decreasing

Correct answer: b

43.  $\text{CaCO}_3 \rightleftharpoons \text{CaO} + \text{CO}_2$  reaches equilibrium in

a) open system

b) closed vessel

c) vacuum

d) high pressure

Correct answer: b

44. Physical equilibrium example

- a)  $\text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{O}(\text{g})$
- b) combustion
- c) displacement
- d) neutralization

Correct answer: a

45. Ionic product water  $K_w$  at 25°C equals

- a)  $10^{-7}$
- b)  $10^{-14}$
- c)  $10^{-1}$
- d) 1

Correct answer: b

46.  $K_w = [\text{H}^+][\text{OH}^-]$  pure water each

- a)  $10^{-14}$  M
- b)  $10^{-7}$  M
- c)  $10^{-1}$  M
- d) 1 M

Correct answer: b

47. pH scale measures

- a)  $\text{H}^+$  concentration
- b)  $\text{OH}^-$  concentration
- c) temperature
- d) pressure

Correct answer: a

48.  $\text{pH} = -\log[\text{H}^+]$  0.01 M  $\text{HNO}_3$  pH equals

- a) 1

b) 2

c) 7

d) 14

Correct answer: b

49. Solution pH 7 considered

a) acidic

b) basic

c) neutral

d) saturated

Correct answer: c

50. Blood pH range

a) 1-3

b) 7.35-7.45

c) 12-14

d) 4-6

Correct answer: b

51. Stomach acid pH approximately

a) 14

b) 7

c) 2

d) 10

Correct answer: c

52. Tooth decay below salivary pH

a) 7.5

b) 5.5

c) 14

d) 1

Correct answer: b

53. Acid rain pH less than

a) 7

b) 5.6

c) 14

d) 10

Correct answer: b

54.  $\text{pH} + \text{pOH}$  equals at  $25^\circ\text{C}$

a) 0

b) 7

c) 14

d) 1

Correct answer: c

55.  $\text{pOH} = 9$  solution pH

a) 5

b) 9

c) 14

d) 23

Correct answer: c

56. Universal indicator used for

a) exact pH

b) approximate pH

c) temperature

d) conductivity

Correct answer: b

57. Milk spoilage summer exemplifies

- a) combination
- b) decomposition
- c) displacement
- d) combustion

Correct answer: b

58. Iron rusting humid environment

- a) combination
- b) decomposition
- c) displacement
- d) double displacement

Correct answer: a

59. Cooking food primarily

- a) physical change
- b) chemical change
- c) reversible
- d) endothermic

Correct answer: b

60. Respiration biological

- a) combination
- b) decomposition
- c) combustion
- d) neutralization

Correct answer: c

61. Word equation zinc sulphuric acid gives

- a)  $\text{ZnSO}_4$  hydrogen

b)  $\text{H}_2\text{SO}_4$  Zn

c) Zn  $\text{H}_2$

d)  $\text{SO}_4$   $\text{H}_2$

Correct answer: a

62. Coefficients chemical equations represent

a) molecules

b) stoichiometric numbers

c) atoms

d) mass

Correct answer: b

63. Calcium oxide silicon dioxide forms

a)  $\text{CaSiO}_3$

b)  $\text{SiO}_2\text{Ca}$

c)  $\text{Ca}_2\text{SiO}_4$

d)  $\text{SiCaO}_2$

Correct answer: a

64. Quicklime water decomposition produces

a)  $\text{CaO}$   $\text{CO}_2$

b)  $\text{Ca}(\text{OH})_2$

c)  $\text{CaCO}_3$

d)  $\text{H}_2\text{O}$

Correct answer: b

65. Zinc copper sulphate produces

a) blue solution

b) reddish brown deposit

c) green precipitate

d) colourless gas

Correct answer: b

66. Lead nitrate potassium iodide yellow

a) gas

b) precipitate

c) solution

d) odour

Correct answer: b

67. Activity series magnesium displaces

a) gold

b) silver

c) sodium

d) all above

Correct answer: d

68. Rate law proportional reactant

a) temperature

b) concentration

c) catalyst

d) surface area

Correct answer: b

69. 10°C rise typically doubles reaction rate for

a) all reactions

b) many reactions

c) gas phase only

d) solid reactions

Correct answer: b

70. Equilibrium constant concentrations

- a) changing
- b) constant
- c) zero
- d) infinite

Correct answer: b

71. Le Chatelier principle predicts equilibrium shift due to

- a) concentration change
- b) temperature change
- c) pressure change
- d) all above

Correct answer: d

72. Soil citrus fruits requires

- a) acidic
- b) alkaline
- c) neutral
- d) saline

Correct answer: b

73. Rice prefers soil pH

- a) acidic
- b) basic
- c) neutral
- d) variable

Correct answer: c

74. pH paper gives readings

- a) exact to 0.01

b) approximate

c) digital

d) continuous

Correct answer: b

75.  $10^{-4}$  M HCl pH equals

a) 3

b) 4

c) 7

d) 10

Correct answer: b

76. pOH 3 solution nature

a) acidic

b) basic

c) neutral

d) saturated

Correct answer: b

77.  $K_w$  increases with

a) decreasing temperature

b) increasing temperature

c) constant temperature

d) pressure

Correct answer: b

78. Buffer solutions maintain

a) changing pH

b) constant pH

c) high pH

d) low pH

Correct answer: b

79. Indicators change colour at specific

a) temperature

b) pH range

c) pressure

d) concentration

Correct answer: b

80. Displacement reaction activity series predicts

a) combination possibility

b) decomposition energy

c) metal reactivity order

d) equilibrium constant

Correct answer: c

81. Double arrow equilibrium indicates

a) irreversible

b) reversible

c) fast reaction

d) slow reaction

Correct answer: b

82. Precipitate insoluble

a) gas

b) liquid

c) solid

d) aqueous

Correct answer: c

83. Combustion always produces

- a) water only
- b) carbon dioxide water
- c) oxygen
- d) hydrogen

Correct answer: b

84. Thermal decomposition endothermic because

- a) bonds broken
- b) bonds formed
- c) gas evolved
- d) precipitate formed

Correct answer: a

85. Electrolytic decomposition requires

- a) heat
- b) light
- c) electricity
- d) catalyst

Correct answer: c

86. Photo decomposition silver chloride produces

- a)  $\text{Ag Cl}_2$
- b)  $\text{Ag} + \text{Cl}_2$
- c)  $2\text{AgCl}$
- d)  $\text{AgCl}_2$

Correct answer: b

87. Reaction rate surface area affects

- a) homogeneous reactions only

b) heterogeneous reactions

c) gas phase only

d) aqueous only

Correct answer: b

88. Catalyst provides pathway

a) higher activation energy

b) lower activation energy

c) same energy

d) no pathway

Correct answer: b

89. Chemical equilibrium dynamic because

a) reactions stop

b) both reactions continue

c) forward stops

b) reverse stops

Correct answer: b

90. Ionic product water constant at constant

a) pressure

b) volume

c) temperature

d) concentration

Correct answer: c

91. pH 14 represents

a) neutral

b) acidic

c) strongly basic

d) weakly basic

Correct answer: c

92. Saliva neutralizes acid pH above

a) 5.5

b) 7

c) 2

d) 14

Correct answer: a

93. Acid rain affects aquatic life by lowering

a) temperature

b) pH

c) oxygen

d) pressure

Correct answer: b

94. pH calculation  $10^{-3}$  M NaOH gives pH

a) 3

b) 11

c) 7

d) 14

Correct answer: b

95. Strong acid complete

a) partial dissociation

b) complete dissociation

c) no dissociation

d) equilibrium

Correct answer: b

96. Weak base solution pH

- a) less than 7
- b) greater than 7 less than 14
- c) exactly 7
- d) exactly 14

Correct answer: b

97. Universal indicator mixture

- a) single dye
- b) multiple dyes
- c) acid
- d) base

Correct answer: b

98. Rate determining step reaction

- a) fastest step
- b) slowest step
- c) equilibrium step
- d) catalytic step

Correct answer: b

99. Order reaction depends on

- a) stoichiometry
- b) experiment
- c) temperature
- d) pressure

Correct answer: b

100. Equilibrium position affected by

- a) adding catalyst

b) changing concentration

c) adding inert gas

d) measuring rate